

MID TERM EXAMINATION

FEBRUARY 2019

Fourth Semester [B. Tech.]
Paper Code : ETMA – 202

Subject : Applied Mathematics-IV
Time : 1 ½ Hrs.
Maximum Marks : 30

Note : Attempt Q.No.1 which is compulsory and any two more questions from remaining. All questions carry equal marks.

Q1 (a) Find the particular integral of $(D-4D')^2 z = 6x \tan(y+4x)$ — (2.5)

(b) Using the method of separation of variables, solve

$$2 \frac{\partial u}{\partial x} + 3 \frac{\partial u}{\partial y} = 0 ; u(0, y) = 2e^{-y} \quad (2.5)$$

(c) A problem in Mathematics is given to three students *A*, *B* and *C* whose chances of solving it are $\frac{1}{2}$, $\frac{3}{4}$ and $\frac{1}{4}$ respectively. What is the probability that the problem would be solved? (2.5)

(d) A random variable *X* has probability density function $f(x) = \begin{cases} e^{-x}, & x \geq 0 \\ 0 & \text{elsewhere} \end{cases}$. Find its moment generating function about origin. (2.5)

Q2 (a) Find the solution of the partial differential equation

$$(D^3 - 4D^2 D' + 4DD'^2)z = 2\cos(3x+2y) \quad (5)$$

(b) Find the solution of the partial differential equation

$$(4D^2 + 3DD' - D'^2 - D - D')z = 3e^{(x+2y)/2} \quad (5)$$

Q3 (a) Three bags contain 2 white, 3 black; 3 white, 2 black; and 4 white and 1 black balls respectively. There is equal probability of each bag being chosen. One ball is drawn from a bag chosen at random. What is the probability that a white ball is drawn? Also, find the probability that it is drawn from the second bag. (5)

(b) The first four moments of a distribution about the value 5 of the variable are 2, 20, 40 and 50 respectively. Find the mean and all the four moments about the mean. Also comment upon skewness and kurtosis. (5)

Q4. A thin uniform tightly stretched vibrating string fixed at the points $x=0$ and $x=l$

satisfies the equation $\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$; $y(x,0) = y_0 \sin^3\left(\frac{\pi x}{l}\right)$ and released from rest from this position. Find the displacement $y(x,t)$ at any x and any time t . (10)

OR

Show that Poisson distribution is a limiting case of Binomial distribution when n is very large and p is small such that np is fixed. Also, using Poisson distribution, find the chance that there will be less than three accidents in a day if 10 accidents took place in a span of 50 days in a town.

(10)

Mid Term Examination
Forth Semester(B.Tech), February-2018
Applied Mathematics-IV(ETMA-202)

Time:1.5 hours

Max.Marks: 30

Note: Attempt three questions including Question number 1 which is compulsory.

1. (i) Solve the partial differential equation $\frac{\partial^3 z}{\partial x^3} - 6 \frac{\partial^3 z}{\partial x^2 \partial y} + 11 \frac{\partial^3 z}{\partial x \partial y^2} - 6 \frac{\partial^3 z}{\partial y^3} = 0$. (2)
- (ii) Use the method of separation of variables to solve the equation $\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial y} + 2u$. (3)
- (iii) Define rth order Moment of random variable X about the point a and find relation between first four order moment of X about mean and origin. (3)
- (iv) A car hire firm has two cars which it hires out day by day. The number of demands for a car on each day is distributed as a Poisson Distribution with mean 1.5. calculate how many days, in a year, car is not used. (2)
2. (a) Solve the linear partial differential equation $2 \frac{\partial^2 z}{\partial x^2} - 5 \frac{\partial^2 z}{\partial x \partial y} + 2 \frac{\partial^2 z}{\partial y^2} = 5 \sin(2x + y)$. (5)
- (b) Solve the partial differential equation $\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial y^2} - 3 \frac{\partial z}{\partial x} + 3 \frac{\partial z}{\partial y} = e^{3x+2y}$. (5)
3. (a) In a normal Distribution, 31% of the items are under 45 and 8% are over 64. Find the mean and standard deviation of the distribution. (5)
- (b) The random variable X can assume the values -1 and 1 with probability $\frac{1}{2}$ each. Find (a) Moment generating function (b) skewness and Kurtosis. (2+3)
4. An insulated rod of length l is initially at a uniform temperature u_0 . Its ends are suddenly cooled to 0°C and kept at that temperature. Find the temperature distribution function $u(x, t)$. (10)

'OR'

Find the constant c such that the function $f(x) = \frac{c}{(x^2+1)}$, where $-\infty < x < \infty$ is a density function of continuous random variable X and compute $P\{\frac{1}{3} \leq X^2 \leq 1\}$ and distribution function $F(X)$. (3 + 4 + 3)

Mid-Term Examination
Fourth Semester [B.Tech.] (Regular)
February-2017

Subject Name: Applied mathematics-IV

Sub.Code: ETMA-202

Time: 1:30 hrs.

Maximum Marks: 30

Note: Q. No. 1 is compulsory and attempts any two from remaining questions. All questions carry equal marks.

1. (a) Find Particular Integral of $(4D^2 + 3DD' - D'^2 - D - D')z = 3e^{(x+2y)/2}$.
- (b) Using the method of separation of variable solve $\frac{\partial u}{\partial x} = 2\frac{\partial u}{\partial y} + u$, where $u(x,0) = e^{-3x} - 2e^{-x}$, $x > 0, y > 0$.
- (c) Find moment generating function of the following random distribution

x	-1	1
P(x)	1/2	1/2
- (d) A can hit a target 3 times in 5 shots, B can hit a target 2 times in 5 shots and C can hit a target 3 times in 4 shots. All of them fire one shot each simultaneously at the target. What is the probability that two shots hit a target.
2. (a) Find complete solution of $(D^2 - 2DD' + D'^2)z = \cos(2x+y)$. (5)
- (b) Solve $\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} - 3\frac{\partial u}{\partial x} + 3\frac{\partial u}{\partial y} = xy$ (5)
3. (a) A die is tossed twice. Getting '5' or '6' on a toss is taken as success. Find the probability distribution, the mean and the variance of the number of successes. (6)
- (b) Find the coefficient Skewness and Kurtosis of a continuous distribution (4)
 with density function defined as $f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} - \infty < x < \infty$
4. Solve the boundary value problem $\frac{\partial^2 y}{\partial t^2} = 4\frac{\partial^2 y}{\partial x^2}$, given that $y(0,t) = 0, y(5,t) = 0, y(x,0) = 0$ and $\left(\frac{\partial y}{\partial t}\right)_{t=0} = \sin(\pi x)$. (10)

'or'

In the normal distribution 7% of the items are under 35 and 89% are 63. Determine the mean and variance of the distribution. Given that $P(0 \leq x \leq 0.18) = 0.07$, $P(0 \leq x \leq 1.48) = 0.43$ and $P(0 \leq x \leq 1.23) = 0.39$. (10)

FIRST TERM EXAMINATION FEBRUARY 2016

Fourth Semester [B. Tech.]
Paper Code : ETMA – 202

Subject : Applied Mathematics-IV
Time : 1:30 Hrs.
Maximum Marks : 30

Note : Attempt Q.No.1 which is compulsory and any two more questions from remaining. All questions carry equal marks.

Q1 (a) Solve $\frac{\partial z}{\partial x} = 2 \frac{\partial z}{\partial y} + y \cos x$. (2.5)

(b) Solve $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial z}{\partial y} = z$. (2.5)

(c) If the probability that the man aged 60 will live till 70 is 0.6, what is the probability that out of 10 men aged 60, 9 men will live upto 70. (2.5)

(d) Determine the value of k , if the probability function of a random variable X is given by

$$p(x) = \begin{cases} \frac{kx}{20}, & x = 1, 2, 3, 4 \\ 0 & \text{for other integers} \end{cases} \quad (2.5)$$

Q2 (a) Find the solution of the partial differential equation

$$(D^3 - 7DD'^2 - 6D'^3)z = \sin(x + 2y). \quad (5)$$

(b) Use the method of separation of variables to solve the partial differential equation

$$4 \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 3u \quad \text{given } u = 3e^{-y} - e^{-5y} \text{ when } x = 0. \quad (5)$$

Q3 (a) Find the solution of the Laplace equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ which satisfies the conditions

$$\begin{aligned} \text{(i) } u \rightarrow 0 \text{ as } y \rightarrow \infty \text{ for all } x & \quad \text{(ii) } u = 0 \text{ at } x = 0 \text{ for all } y \\ \text{(iii) } u = 0 \text{ at } x = l \text{ for all } y & \quad \text{(iv) } u = lx - x^2 \text{ if } y = 0 \text{ for all } x \in (0, l). \end{aligned} \quad (5)$$

(b) An insurance company insured 2000 scooter drivers, 4000 car drivers and 6000 truck drivers. The probability of an accident involving a scooter driver, a car driver and a truck driver is 0.01, 0.03 and 0.15 respectively. One of the insured person meets with an accident. What is the probability that he is a scooter driver. (5)

Q4 (a) Find the moment generating function of the distribution $f(x) = \frac{1}{c} e^{-x/c}$, $0 \leq x < \infty$, $c > 0$ about origin. Hence find its mean and standard deviation. (5)

(b) A manufacturer of pins knows that on an average 5% of his product is defective. He sells pins in boxes of 100 and guarantees that not more than 4 pins will be defective. What is the probability that the box will fail to meet the guaranteed quality? ($e^{-5} = 0.0067$). (5)

(Please write your Roll No. immediately)

Roll No.

Second Term Examination
APRIL 2016

Fourth Semester [B. Tech.]
Paper Code : ETMA – 202

Subject : Applied Mathematics-IV
Time : 1:30 Hour
Maximum Marks : 30

Note: Attempt Q.1 which is compulsory and any two more questions from remaining.
All questions carry equal marks.

Q1 (a) Prove that $r = \frac{\sigma_x^2 + \sigma_y^2 - \sigma_{x-y}^2}{2\sigma_x\sigma_y}$, where σ_x & σ_y are the S.D.'s of x and y -series respectively and r is the correlation coefficient. (2.5)

(b) A random sample of 900 measurements from a large population gave a mean value of 64. If this sample has been drawn from a normal population with standard deviation of 20, find the 95% confidence limits for the mean in the population. (2.5)

(c) A toy company manufactures two types of toys, type A and type B. Each toy of type B takes twice as long to produce as one of type A, and the company would have time to make a maximum of 2000 toys per day. The supply of plastic is sufficient to produce 1500 toys per day (both A and B combined). Type B toy requires a dress of which there are only 600 per day available. If the company makes a profit of Rs.3.00 and Rs.5.00 per toy, respectively on type A and B, then how many of each types of toy should be produced per day in order to maximize the total profit. Formulate this problem. (2.5)

(d) Write the dual of the following problem: $\text{Max } Z = 3x_1 + 2x_2$
s.t. $x_1 - x_2 \leq 1$, $x_1 + x_2 \geq 3$, $x_1 \geq 0$, x_2 is unrestricted in sign. (2.5)

Q2 (a) By the method of least squares, fit a parabola from the following data:

x:	1	2	3	4	5
y:	2	6	4	5	2

 (5)

(b) The equations of two lines of regression are $4x + 3y + 7 = 0$ and $3x + 4y + 8 = 0$. Find the regression coefficients b_{yx} , b_{xy} and the correlation coefficient r . Also, find the standard deviation of y , if the variance of x is 4. (5)

Q3 (a) A random sample of 10 boys had the following IQ:

70, 120, 110, 101, 88, 83, 95, 98, 107, 100.

Do these data support the assumption of population mean IQ of 100 at 5% level of significance? (Given $t_{0.05}=2.26$ for 9 d.f., $t_{0.05}=2.23$ for 10 d.f., $t_{0.05}=2.20$ for 11 d.f.) (5)

(b) Solve the following L.P.P. $\text{Max } Z = -2x_1 - x_3$ s.t.

$x_1 + x_2 - x_3 \geq 5$, $x_1 - 2x_2 + 4x_3 \geq 8$, and $x_1, x_2, x_3 \geq 0$. (5)

Q4 (a) A car hire company has one car at each of five depots a, b, c, d and e . A customer requires a car in each town, namely A, B, C, D and E . Distances (in kms) between depots(origins) and towns (destinations) are given in the following distance matrix:

P.T.O.

FIRST TERM EXAMINATION
FEBRUARY 2015

Fourth Semester [B. Tech.]
Paper Code : ETMA – 202

Subject : Applied Mathematics-IV
Time : 1 Hour
Maximum Marks : 30

Note : Attempt Q.No.1 which is compulsory and any two more questions from remaining. All questions carry equal marks.

Q1 (a) Solve $r - s - 2t = (y - 1)e^x$. (2.5)

(b) Solve $\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u$; $u(x, 0) = 6e^{-3x}$, $x > 0$. (2.5)

(c) The odds against A solving a certain problem are 5 to 7 and the odds in favour of B solving the same problem are 3 to 4. What is the probability that if both of them try, the problem would be solved. (2.5)

(d) A random variable X has probability function $p(x) = \frac{1}{2^x}$, $x = 1, 2, 3, \dots$. Find its moment generating function about origin. (2.5)

Q2 (a) Find the solution of the partial differential equation

$$(D - 3D' - 2)^2 z = 2xe^{2x} \tan(y + 3x) \quad (5)$$

(b) A rod of length l with insulated sides is initially at a uniform temperature u_0 . Its ends are suddenly cooled to 0°C and are kept at that temperature. Find the temperature function $u(x, t)$. (5)

Q3 (a) Solve the differential equation $\frac{\partial^2 u}{\partial t^2} = 4 \frac{\partial^2 u}{\partial x^2}$ subject to the conditions $u = \sin t$ at

$$x = 0 \text{ and } \frac{\partial u}{\partial x} = \sin t \text{ at } x = 0. \quad (5)$$

(b) In a toy factory, machines A , B and C manufacture respectively 25%, 35% and 40% of the total. Of their output 5, 4, 2 percents are respectively defective. A toy is drawn at random from the total production. What is the probability that the toy drawn is defective? Also find the probability that it was manufactured by machine A . (5)

Q4 (a) The first four moments of a distribution about the value 4 of the variable are 1, 4, 10 and 45 respectively. Find the mean and all the four moments about the mean. Also comment upon skewness and kurtosis. (5)

(b) X is a continuous random variable with probability density function given by

$$f(x) = \begin{cases} kx & ; 0 \leq x < 5 \\ k(10 - x) & ; 5 \leq x < 10 \\ 0 & ; \text{otherwise} \end{cases}$$

(i) Find the value of k (ii) Mean of X (iii) $P(5 < X \leq 12)$. (2+2+1)